

DANDI:000021: Allen Institute - Visual Coding - Neuropixels (Brain Observatory 1.1 Stimulus Set)

Comparison with the Allen Brain Observatory's own tables (computed by this export; files in original_files/allen_ecephys_cache): across 000021 + 000022 the NWB files contain 136,556 units, and Allen's units.csv lists 135,275, all of which are in the NWB files. 1,281 NWB units in 50 sessions are not in units.csv. On the 135,275 shared units, the NWB electrode location and Allen's channels.csv ecephys_structure_acronym agree for 99.0%.

Scope of this export (computed by this export): the units, electrodes and intervals tables of every NWB file (session files; the per-probe LFP files were skipped) were read remotely from the DANDI Archive; voltage, LFP, video and other time series were not downloaded.

Source labels: **DANDI metadata** = the Dandiset's published metadata; **NWB file** = values or text stored in the NWB files; **Computed by this export** = derived by the export scripts in dandi_export/ (definitions given where used).

1. The dataset as published

Everything in this section is quoted from the Dandiset's published metadata or, for file-level fields, from sub-699733573_ses-715093703.nwb.

Field	Value (DANDI metadata)
Name	Allen Institute - Visual Coding - Neuropixels (Brain Observatory 1.1 Stimulus Set)
Identifier / version	DANDI:000021 / 0.251116.2246
Citation	Siegle, Josh; Wakeman, Wayne; Jia, Xiaoxuan; Heller, Gregg; Ramirez, Tamina; Graddis, Nile; Mei, Nicholas; Durand, Severine (2025) Allen Institute - Visual Coding - Neuropixels (Brain Observatory 1.1 Stimulus Set) (Version 0.251116.2246) [Data set]. DANDI Archive. https://doi.org/10.48324/dandi.000021/0.251116.2246
DOI	10.48324/dandi.000021/0.251116.2246
Contributors (roles)	Siegle, Josh (Author, ContactPerson, ProjectLeader); Wakeman, Wayne (DataManager, ContactPerson); Jia, Xiaoxuan (Validation); Heller, Gregg (DataCollector); Ramirez, Tamina (DataCollector); Graddis, Nile (Software); Mei, Nicholas (Software); Durand, Severine (DataCollector)
License	CC-BY-4.0
Keywords	electrophysiology, life sciences, machine learning, neurobiology, signal processing
Files / size / subjects	214 files, 477.56 GB, 32 subjects
Species	House mouse
Approach	electrophysiological approach
Measurement technique	analytical technique; signal filtering technique; spike sorting technique
Variables measured	ProcessingModule, LFP, Units
Date published	2025-11-16T22:46:42.638124+00:00

Description (DANDI metadata, verbatim)

Allen Institute October 2019 Mouse extracellular electrophysiology data approximately matching two-photon brain observatory stimulus set.

For more information, see <https://portal.brain-map.org/explore/circuits/visual-coding-neuropixels>

Data are subject to Allen Institute terms of use, available at: <http://www.alleninstitute.org/legal/terms-use/>

File-level fields (NWB file, verbatim)

Field	Value (NWB file)
session_description	Data and metadata for an Ecephys session

Field	Value (NWB file)
institution	Allen Institute for Brain Science
stimulus_notes	brain_observatory_1.1
subject	subject_id: 699733573; species: Mus musculus; genotype: Sst-IRES-Cre/wt;Ai32(RCL-ChR2(H134R)_EYFP)/wt; sex: M; age: P118D
electrode_groups	probeA: description = 'Ecephys Electrode Group', location = 'See electrode locations'; probeB: description = 'Ecephys Electrode Group', location = 'See electrode locations'; probeC: description = 'Ecephys Electrode Group', location = 'See electrode locations'; probeD: description = 'Ecephys Electrode Group', location = 'See electrode locations'; probeE: description = 'Ecephys Electrode Group', location = 'See electrode locations'; probeF: description = 'Ecephys Electrode Group', location = 'See electrode locations'

2. Contents of the files

Quantity	Value	Source
NWB files read / subjects	32 / 32	NWB file
Species	Mus musculus (32)	NWB file
Sex (files)	M (27), F (5)	NWB file
Genotype (files)	wt/wt (16), Sst-IRES-Cre/wt;Ai32(RCL-ChR2(H134R)_EYFP)/wt (6), Pvalb-IRES-Cre/wt;Ai32(RCL-ChR2(H134R)_EYFP)/wt (5), Vip-IRES-Cre/wt;Ai32(RCL-ChR2(H134R)_EYFP)/wt (5)	NWB file
Strain (files)		NWB file
Age (files)	P100D (3), P110D (3), P122D (2), P118D (1), P125D (1), P112D (1), P108D (1), P140D (1), P121D (1), P120D (1), P91D (1), P92D (1), P98D (1), P141D (1), P106D (1), P93D (1), P105D (1), P96D (1), P102D (1), P103D (1), P113D (1), P126D (1), P119D (1), P124D (1), P116D (1), P129D (1), P107D (1)	NWB file
Lab / institution	/ Allen Institute for Brain Science (32)	NWB file
Units (total)	74,658	NWB file
Units per file	median 2292 (range 1603–3232)	NWB file
Spikes (sum of spike_times_index lengths)		NWB file
How region was assigned to units	electrodes.location at units.peak_channel_id (electrode id)	Computed by this export
Distinct region labels	49	NWB file
units.quality values	good: 54,476, noise: 20,182	NWB file

The meaning of each units column, including any quality label, is quoted from the NWB file in section 7.

3. NWB files

File (session_id)	Subject	Sex	Age	Size (GB)	Units	Electrodes	Electrode groups	Interval tables (rows)
sub-699733573_ses-715093703	699733573	M	P118D	2.86	2779	2304	probeA, probeB, probeC, probeD, probeE, probeF	drifting_gratings_presentations(628); flashes_presentations(150); gabors_presentations(3645); invalid_times(8); natural_movie_one_presentations(18000); natural_movie_three_presentations(36000); natural_scenes_presentations(5950); spontaneous_presentations(15); static_gratings_presentations(6000)
sub-703279277_ses-719161530	703279277	M	P122D	3.07	3232	2304	probeA, probeB, probeC, probeD, probeE, probeF	drifting_gratings_presentations(628); flashes_presentations(150); gabors_presentations(3645); invalid_times(1); natural_movie_one_presentations(18000); natural_movie_three_presentations(36000); natural_scenes_presentations(5950); spontaneous_presentations(15); static_gratings_presentations(6000)
sub-707296975_ses-721123822	707296975	M	P125D	1.74	1603	2304	probeA, probeB, probeC, probeD, probeE, probeF	drifting_gratings_presentations(628); flashes_presentations(150); gabors_presentations(3645); invalid_times(5); natural_movie_one_presentations(18000); natural_movie_three_presentations(36000); natural_scenes_presentations(5950); spontaneous_presentations(15); static_gratings_presentations(6000)
sub-716813540_ses-739448407	716813540	M	P112D	2.16	2370	2304	probeA, probeB, probeC, probeD, probeE, probeF	drifting_gratings_presentations(630); flashes_presentations(150); gabors_presentations(3645); invalid_times(4); natural_movie_one_presentations(18000); natural_movie_three_presentations(36000); natural_scenes_presentations(5950); spontaneous_presentations(15); static_gratings_presentations(6000)
sub-717038285_ses-732592105	717038285	M	P100D	2.91	3076	1920	probeB, probeC, probeD, probeE, probeF	drifting_gratings_presentations(628); flashes_presentations(150); gabors_presentations(3645); invalid_times(4); natural_movie_one_presentations(18000); natural_movie_three_presentations(36000); natural_scenes_presentations(5950); spontaneous_presentations(15); static_gratings_presentations(6000)
sub-718643564_ses-737581020	718643564	M	P108D	2.18	2055	2304	probeA, probeB, probeC, probeD, probeE, probeF	drifting_gratings_presentations(630); flashes_presentations(150); gabors_presentations(3645); invalid_times(9); natural_movie_one_presentations(18000); natural_movie_three_presentations(36000); natural_scenes_presentations(5950); spontaneous_presentations(15); static_gratings_presentations(6000)
sub-719817799_ses-744228101	719817799	M	P122D	2.65	2695	2304	probeA, probeB, probeC, probeD, probeE, probeF	drifting_gratings_presentations(630); flashes_presentations(150); gabors_presentations(3645); natural_movie_one_presentations(18000); natural_movie_three_presentations(36000); natural_scenes_presentations(5950); spontaneous_presentations(15); static_gratings_presentations(6000)
sub-719828686_ses-754312389	719828686	M	P140D	1.86	1891	2304	probeA, probeB, probeC, probeD, probeE, probeF	drifting_gratings_presentations(630); flashes_presentations(150); gabors_presentations(3645); invalid_times(1); natural_movie_one_presentations(18000); natural_movie_three_presentations(36000); natural_scenes_presentations(5950); spontaneous_presentations(15); static_gratings_presentations(6000)

File (session_id)	Subject	Sex	Age	Size (GB)	Units	Electrodes	Electrode groups	Interval tables (rows)
sub-722882751_ses-743475441	722882751	M	P121D	2.12	2133	2304	probeA, probeB, probeC, probeD, probeE, probeF	drifting_gratings_presentations(630); flashes_presentations(150); gabors_presentations(3645); invalid_times(8); natural_movie_one_presentations(18000); natural_movie_three_presentations(36000); natural_scenes_presentations(5950); spontaneous_presentations(15); static_gratings_presentations(6000)
sub-723627600_ses-742951821	723627600	M	P120D	2.92	3008	2304	probeA, probeB, probeC, probeD, probeE, probeF	drifting_gratings_presentations(630); flashes_presentations(150); gabors_presentations(3645); invalid_times(7); natural_movie_one_presentations(18000); natural_movie_three_presentations(36000); natural_scenes_presentations(5950); spontaneous_presentations(15); static_gratings_presentations(6000)
sub-726141242_ses-750332458	726141242	M	P91D	3.24	3047	2304	probeA, probeB, probeC, probeD, probeE, probeF	drifting_gratings_presentations(630); flashes_presentations(150); gabors_presentations(3645); natural_movie_one_presentations(18000); natural_movie_three_presentations(36000); natural_scenes_presentations(5950); spontaneous_presentations(15); static_gratings_presentations(6000)
sub-726162193_ses-750749662	726162193	M	P92D	2.82	2489	2304	probeA, probeB, probeC, probeD, probeE, probeF	drifting_gratings_presentations(630); flashes_presentations(150); gabors_presentations(3645); natural_movie_one_presentations(18000); natural_movie_three_presentations(36000); natural_scenes_presentations(5950); spontaneous_presentations(15); static_gratings_presentations(6000)
sub-726170927_ses-746083955	726170927	F	P98D	2.56	2098	2304	probeA, probeB, probeC, probeD, probeE, probeF	drifting_gratings_presentations(630); flashes_presentations(150); gabors_presentations(3645); natural_movie_one_presentations(18000); natural_movie_three_presentations(36000); natural_scenes_presentations(5950); spontaneous_presentations(15); static_gratings_presentations(6000)
sub-726298249_ses-754829445	726298249	M	P141D	2.70	2299	1920	probeA, probeB, probeC, probeE, probeF	drifting_gratings_presentations(630); flashes_presentations(150); gabors_presentations(3645); natural_movie_one_presentations(18000); natural_movie_three_presentations(36000); natural_scenes_presentations(5950); spontaneous_presentations(15); static_gratings_presentations(6000)
sub-730756767_ses-757970808	730756767	M	P106D	2.38	2422	1920	probeA, probeC, probeD, probeE, probeF	drifting_gratings_presentations(630); flashes_presentations(150); gabors_presentations(3645); natural_movie_one_presentations(18000); natural_movie_three_presentations(36000); natural_scenes_presentations(5950); spontaneous_presentations(15); static_gratings_presentations(6000)
sub-730760263_ses-755434585	730760263	M	P100D	2.24	2127	2304	probeA, probeB, probeC, probeD, probeE, probeF	drifting_gratings_presentations(630); flashes_presentations(150); gabors_presentations(3645); natural_movie_one_presentations(18000); natural_movie_three_presentations(36000); natural_scenes_presentations(5950); spontaneous_presentations(15); static_gratings_presentations(6000)
sub-732548371_ses-751348571	732548371	F	P93D	3.07	2645	2304	probeA, probeB, probeC, probeD, probeE, probeF	drifting_gratings_presentations(630); flashes_presentations(150); gabors_presentations(3645); natural_movie_one_presentations(18000); natural_movie_three_presentations(36000); natural_scenes_presentations(5950); spontaneous_presentations(15); static_gratings_presentations(6000)
sub-733457986_ses-757216464	733457986	M	P105D	3.04	2812	2304	probeA, probeB, probeC, probeD, probeE, probeF	drifting_gratings_presentations(630); flashes_presentations(150); gabors_presentations(3645); invalid_times(1); natural_movie_one_presentations(18000); natural_movie_three_presentations(36000); natural_scenes_presentations(5950); spontaneous_presentations(15); static_gratings_presentations(6000)

File (session_id)	Subject	Sex	Age	Size (GB)	Units	Electrodes	Electrode groups	Interval tables (rows)
sub-734865729_ses-756029989	734865729	M	P96D	2.58	2169	2304	probeA, probeB, probeC, probeD, probeE, probeF	drifting_gratings_presentations(630); flashes_presentations(150); gabors_presentations(3645); invalid_times(1); natural_movie_one_presentations(18000); natural_movie_three_presentations(36000); natural_scenes_presentations(5950); spontaneous_presentations(15); static_gratings_presentations(6000)
sub-735109599_ses-758798717	735109599	M	P102D	2.20	1702	1536	probeA, probeB, probeC, probeE	drifting_gratings_presentations(630); flashes_presentations(150); gabors_presentations(3645); invalid_times(1); natural_movie_one_presentations(18000); natural_movie_three_presentations(36000); natural_scenes_presentations(5950); spontaneous_presentations(15); static_gratings_presentations(6000)
sub-738651046_ses-760693773	738651046	F	P110D	2.86	2285	2304	probeA, probeB, probeC, probeD, probeE, probeF	drifting_gratings_presentations(630); flashes_presentations(150); gabors_presentations(3645); natural_movie_one_presentations(18000); natural_movie_three_presentations(36000); natural_scenes_presentations(5950); spontaneous_presentations(15); static_gratings_presentations(6000)
sub-739783158_ses-760345702	739783158	M	P103D	1.96	1784	1920	probeA, probeB, probeC, probeD, probeE	drifting_gratings_presentations(630); flashes_presentations(150); gabors_presentations(3645); natural_movie_one_presentations(18000); natural_movie_three_presentations(36000); natural_scenes_presentations(5950); spontaneous_presentations(15); static_gratings_presentations(6000)
sub-740268983_ses-759883607	740268983	M	P113D	1.86	1763	1536	probeA, probeB, probeC, probeE	drifting_gratings_presentations(630); flashes_presentations(150); gabors_presentations(3645); natural_movie_one_presentations(18000); natural_movie_three_presentations(36000); natural_scenes_presentations(5950); spontaneous_presentations(15); static_gratings_presentations(6000)
sub-742602884_ses-763673393	742602884	M	P126D	2.54	2455	2304	probeA, probeB, probeC, probeD, probeE, probeF	drifting_gratings_contrast_presentations(540); drifting_gratings_presentations(630); flashes_presentations(150); gabors_presentations(3645); natural_movie_one_presentations(18000); natural_movie_three_presentations(36000); natural_scenes_presentations(5950); spontaneous_presentations(16); static_gratings_presentations(6000)
sub-742714472_ses-761418226	742714472	M	P119D	2.67	2371	2304	probeA, probeB, probeC, probeD, probeE, probeF	drifting_gratings_contrast_presentations(540); drifting_gratings_presentations(630); flashes_presentations(150); gabors_presentations(3645); natural_movie_one_presentations(18000); natural_movie_three_presentations(36000); natural_scenes_presentations(5950); spontaneous_presentations(16); static_gratings_presentations(6000)
sub-744915196_ses-762602078	744915196	M	P110D	2.02	2159	2304	probeA, probeB, probeC, probeD, probeE, probeF	drifting_gratings_contrast_presentations(540); drifting_gratings_presentations(630); flashes_presentations(150); gabors_presentations(3645); natural_movie_one_presentations(18000); natural_movie_three_presentations(36000); natural_scenes_presentations(5950); spontaneous_presentations(16); static_gratings_presentations(6000)
sub-745276220_ses-762120172	745276220	M	P100D	2.87	2361	1920	probeA, probeB, probeC, probeD, probeE	drifting_gratings_contrast_presentations(540); drifting_gratings_presentations(630); flashes_presentations(150); gabors_presentations(3645); natural_movie_one_presentations(18000); natural_movie_three_presentations(36000); natural_scenes_presentations(5950); spontaneous_presentations(16); static_gratings_presentations(6000)

File (session_id)	Subject	Sex	Age	Size (GB)	Units	Electrodes	Electrode groups	Interval tables (rows)
sub-757329617_ses-773418906	757329617	F	P124D	1.93	1880	2304	probeA, probeB, probeC, probeD, probeE, probeF	drifting_gratings_presentations(630); flashes_presentations(150); gabors_presentations(3645); natural_movie_one_presentations(18000); natural_movie_three_presentations(36000); natural_scenes_presentations(5950); spontaneous_presentations(15); static_gratings_presentations(6000)
sub-769360771_ses-791319847	769360771	M	P116D	2.32	2020	2304	probeA, probeB, probeC, probeD, probeE, probeF	drifting_gratings_contrast_presentations(540); drifting_gratings_presentations(630); flashes_presentations(150); gabors_presentations(3645); natural_movie_one_presentations(18000); natural_movie_three_presentations(36000); natural_scenes_presentations(5950); spontaneous_presentations(16); static_gratings_presentations(6000)
sub-772616820_ses-799864342	772616820	M	P129D	2.74	2249	2304	probeA, probeB, probeC, probeD, probeE, probeF	drifting_gratings_contrast_presentations(540); drifting_gratings_presentations(630); flashes_presentations(150); gabors_presentations(3645); natural_movie_one_presentations(18000); natural_movie_three_presentations(36000); natural_scenes_presentations(5950); spontaneous_presentations(16); static_gratings_presentations(6000)
sub-775876808_ses-798911424	775876808	F	P110D	2.86	2525	2304	probeA, probeB, probeC, probeD, probeE, probeF	drifting_gratings_contrast_presentations(540); drifting_gratings_presentations(630); flashes_presentations(150); gabors_presentations(3645); natural_movie_one_presentations(18000); natural_movie_three_presentations(36000); natural_scenes_presentations(5950); spontaneous_presentations(16); static_gratings_presentations(6000)
sub-776061236_ses-797828357	776061236	M	P107D	2.55	2154	2304	probeA, probeB, probeC, probeD, probeE, probeF	drifting_gratings_contrast_presentations(540); drifting_gratings_presentations(630); flashes_presentations(150); gabors_presentations(3645); invalid_times(4); natural_movie_one_presentations(18000); natural_movie_three_presentations(36000); natural_scenes_presentations(5950); spontaneous_presentations(16); static_gratings_presentations(6000)

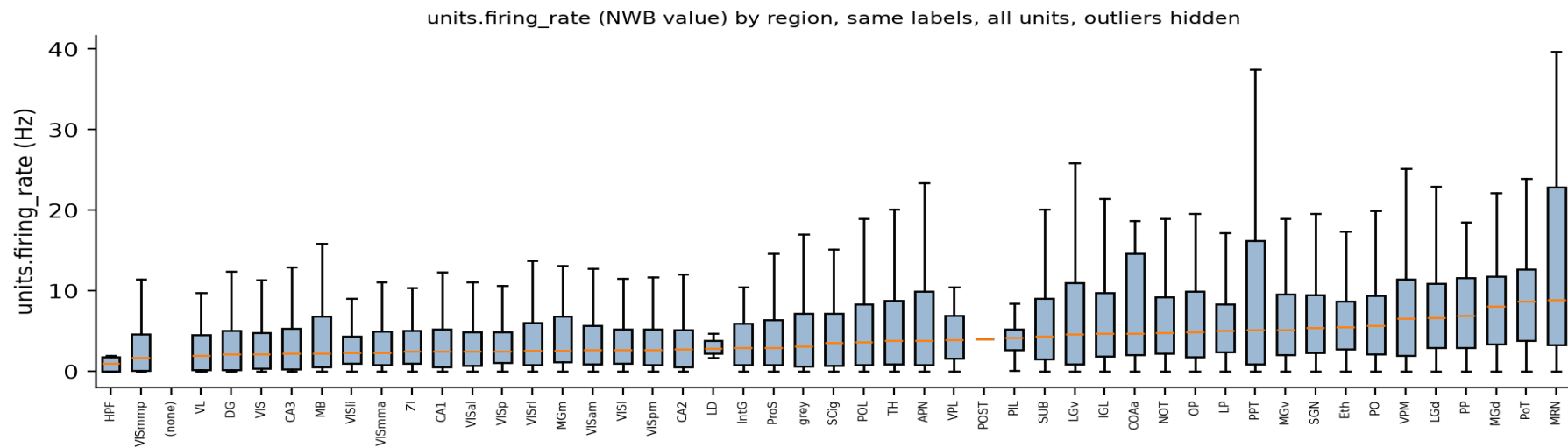
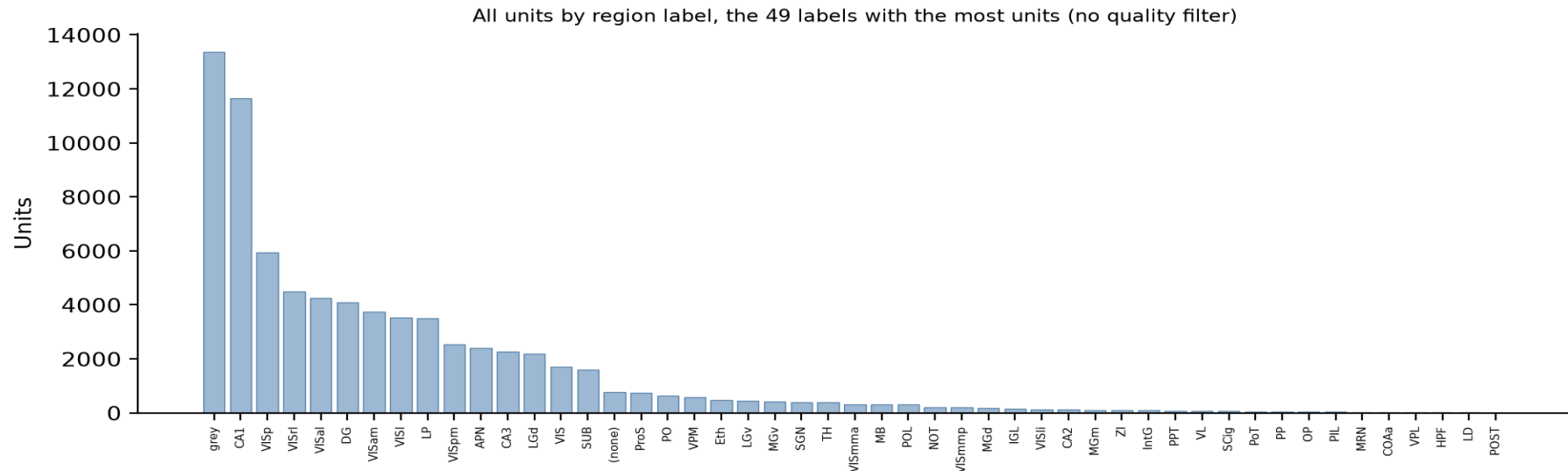
Other data in each file (not exported)

File (session_id)	Acquisition	Processing modules: interfaces
sub-699733573_ses-715093703	raw_running_wheel_rotation, running_wheel_signal_voltage, running_wheel_supply_voltage	eye_tracking_rig_metadata:eye_tracking_rig_metadata; optotagging:optotagging/optogenetic_stimulation; running:running_speed/running_speed_end_times/running_wheel_rotation; stimulus:timestamps
sub-703279277_ses-719161530	raw_running_wheel_rotation, running_wheel_signal_voltage, running_wheel_supply_voltage	eye_tracking_rig_metadata:eye_tracking_rig_metadata; optotagging:optotagging/optogenetic_stimulation; running:running_speed/running_speed_end_times/running_wheel_rotation; stimulus:timestamps
sub-707296975_ses-721123822	raw_running_wheel_rotation, running_wheel_signal_voltage, running_wheel_supply_voltage	eye_tracking_rig_metadata:eye_tracking_rig_metadata; optotagging:optotagging/optogenetic_stimulation; running:running_speed/running_speed_end_times/running_wheel_rotation; stimulus:timestamps
sub-716813540_ses-739448407	raw_running_wheel_rotation, running_wheel_signal_voltage, running_wheel_supply_voltage	eye_tracking_rig_metadata:eye_tracking_rig_metadata; running:running_speed/running_speed_end_times/running_wheel_rotation; stimulus:timestamps
sub-717038285_ses-732592105	raw_running_wheel_rotation, running_wheel_signal_voltage, running_wheel_supply_voltage	eye_tracking_rig_metadata:eye_tracking_rig_metadata; running:running_speed/running_speed_end_times/running_wheel_rotation; stimulus:timestamps
sub-718643564_ses-737581020	raw_running_wheel_rotation, running_wheel_signal_voltage, running_wheel_supply_voltage	eye_tracking_rig_metadata:eye_tracking_rig_metadata; running:running_speed/running_speed_end_times/running_wheel_rotation; stimulus:timestamps

File (session_id)	Acquisition	Processing modules: interfaces
sub-733457986_ses-757216464	raw_running_wheel_rotation, running_wheel_signal_voltage, running_wheel_supply_voltage	eye_tracking:cr_ellipse_fits/eye_ellipse_fits/pupil_ellipse_fits; eye_tracking_rig_metadata:eye_tracking_rig_metadata; filtered_gaze_mapping:eye_area/pupil_area/screen_coordinates/screen_coordinates_spherical; optotagging:optotagging/optogenetic_stimulation; raw_gaze_mapping:eye_area/pupil_area/screen_coordinates/screen_coordinates_spherical; running:running_speed/running_speed_end_times/running_wheel_rotation; stimulus:timestamps
sub-734865729_ses-756029989	raw_running_wheel_rotation, running_wheel_signal_voltage, running_wheel_supply_voltage	eye_tracking:cr_ellipse_fits/eye_ellipse_fits/pupil_ellipse_fits; eye_tracking_rig_metadata:eye_tracking_rig_metadata; filtered_gaze_mapping:eye_area/pupil_area/screen_coordinates/screen_coordinates_spherical; optotagging:optotagging/optogenetic_stimulation; raw_gaze_mapping:eye_area/pupil_area/screen_coordinates/screen_coordinates_spherical; running:running_speed/running_speed_end_times/running_wheel_rotation; stimulus:timestamps
sub-735109599_ses-758798717	raw_running_wheel_rotation, running_wheel_signal_voltage, running_wheel_supply_voltage	eye_tracking:cr_ellipse_fits/eye_ellipse_fits/pupil_ellipse_fits; eye_tracking_rig_metadata:eye_tracking_rig_metadata; filtered_gaze_mapping:eye_area/pupil_area/screen_coordinates/screen_coordinates_spherical; optotagging:optotagging/optogenetic_stimulation; raw_gaze_mapping:eye_area/pupil_area/screen_coordinates/screen_coordinates_spherical; running:running_speed/running_speed_end_times/running_wheel_rotation; stimulus:timestamps
sub-738651046_ses-760693773	raw_running_wheel_rotation, running_wheel_signal_voltage, running_wheel_supply_voltage	eye_tracking:cr_ellipse_fits/eye_ellipse_fits/pupil_ellipse_fits; eye_tracking_rig_metadata:eye_tracking_rig_metadata; filtered_gaze_mapping:eye_area/pupil_area/screen_coordinates/screen_coordinates_spherical; optotagging:optotagging/optogenetic_stimulation; raw_gaze_mapping:eye_area/pupil_area/screen_coordinates/screen_coordinates_spherical; running:running_speed/running_speed_end_times/running_wheel_rotation; stimulus:timestamps
sub-739783158_ses-760345702	raw_running_wheel_rotation, running_wheel_signal_voltage, running_wheel_supply_voltage	eye_tracking:cr_ellipse_fits/eye_ellipse_fits/pupil_ellipse_fits; eye_tracking_rig_metadata:eye_tracking_rig_metadata; filtered_gaze_mapping:eye_area/pupil_area/screen_coordinates/screen_coordinates_spherical; optotagging:optotagging/optogenetic_stimulation; raw_gaze_mapping:eye_area/pupil_area/screen_coordinates/screen_coordinates_spherical; running:running_speed/running_speed_end_times/running_wheel_rotation; stimulus:timestamps
sub-740268983_ses-759883607	raw_running_wheel_rotation, running_wheel_signal_voltage, running_wheel_supply_voltage	eye_tracking:cr_ellipse_fits/eye_ellipse_fits/pupil_ellipse_fits; eye_tracking_rig_metadata:eye_tracking_rig_metadata; filtered_gaze_mapping:eye_area/pupil_area/screen_coordinates/screen_coordinates_spherical; optotagging:optotagging/optogenetic_stimulation; raw_gaze_mapping:eye_area/pupil_area/screen_coordinates/screen_coordinates_spherical; running:running_speed/running_speed_end_times/running_wheel_rotation; stimulus:timestamps
sub-742602884_ses-763673393	raw_running_wheel_rotation, running_wheel_signal_voltage, running_wheel_supply_voltage	eye_tracking:cr_ellipse_fits/eye_ellipse_fits/pupil_ellipse_fits; eye_tracking_rig_metadata:eye_tracking_rig_metadata; filtered_gaze_mapping:eye_area/pupil_area/screen_coordinates/screen_coordinates_spherical; optotagging:optotagging/optogenetic_stimulation; raw_gaze_mapping:eye_area/pupil_area/screen_coordinates/screen_coordinates_spherical; running:running_speed/running_speed_end_times/running_wheel_rotation; stimulus:timestamps
sub-742714472_ses-761418226	raw_running_wheel_rotation, running_wheel_signal_voltage, running_wheel_supply_voltage	eye_tracking:cr_ellipse_fits/eye_ellipse_fits/pupil_ellipse_fits; eye_tracking_rig_metadata:eye_tracking_rig_metadata; filtered_gaze_mapping:eye_area/pupil_area/screen_coordinates/screen_coordinates_spherical; optotagging:optotagging/optogenetic_stimulation; raw_gaze_mapping:eye_area/pupil_area/screen_coordinates/screen_coordinates_spherical; running:running_speed/running_speed_end_times/running_wheel_rotation; stimulus:timestamps
sub-744915196_ses-762602078	raw_running_wheel_rotation, running_wheel_signal_voltage, running_wheel_supply_voltage	eye_tracking:cr_ellipse_fits/eye_ellipse_fits/pupil_ellipse_fits; eye_tracking_rig_metadata:eye_tracking_rig_metadata; filtered_gaze_mapping:eye_area/pupil_area/screen_coordinates/screen_coordinates_spherical; optotagging:optotagging/optogenetic_stimulation; raw_gaze_mapping:eye_area/pupil_area/screen_coordinates/screen_coordinates_spherical; running:running_speed/running_speed_end_times/running_wheel_rotation; stimulus:timestamps
sub-745276220_ses-762120172	raw_running_wheel_rotation, running_wheel_signal_voltage, running_wheel_supply_voltage	eye_tracking:cr_ellipse_fits/eye_ellipse_fits/pupil_ellipse_fits; eye_tracking_rig_metadata:eye_tracking_rig_metadata; filtered_gaze_mapping:eye_area/pupil_area/screen_coordinates/screen_coordinates_spherical; optotagging:optotagging/optogenetic_stimulation; raw_gaze_mapping:eye_area/pupil_area/screen_coordinates/screen_coordinates_spherical; running:running_speed/running_speed_end_times/running_wheel_rotation; stimulus:timestamps

File (session_id)	Acquisition	Processing modules: interfaces
sub-757329617_ses-773418906	raw_running_wheel_rotation, running_wheel_signal_voltage, running_wheel_supply_voltage	eye_tracking:cr_ellipse_fits/eye_ellipse_fits/pupil_ellipse_fits; eye_tracking_rig_metadata:eye_tracking_rig_metadata; filtered_gaze_mapping:eye_area/pupil_area/screen_coordinates/screen_coordinates_spherical; optotagging:optotagging/optogenetic_stimulation; raw_gaze_mapping:eye_area/pupil_area/screen_coordinates/screen_coordinates_spherical; running:running_speed/running_speed_end_times/running_wheel_rotation; stimulus:timestamps
sub-769360771_ses-791319847	raw_running_wheel_rotation, running_wheel_signal_voltage, running_wheel_supply_voltage	eye_tracking:cr_ellipse_fits/eye_ellipse_fits/pupil_ellipse_fits; eye_tracking_rig_metadata:eye_tracking_rig_metadata; filtered_gaze_mapping:eye_area/pupil_area/screen_coordinates/screen_coordinates_spherical; optotagging:optotagging/optogenetic_stimulation; raw_gaze_mapping:eye_area/pupil_area/screen_coordinates/screen_coordinates_spherical; running:running_speed/running_speed_end_times/running_wheel_rotation; stimulus:timestamps
sub-772616820_ses-799864342	raw_running_wheel_rotation, running_wheel_signal_voltage, running_wheel_supply_voltage	eye_tracking:cr_ellipse_fits/eye_ellipse_fits/pupil_ellipse_fits; eye_tracking_rig_metadata:eye_tracking_rig_metadata; filtered_gaze_mapping:eye_area/pupil_area/screen_coordinates/screen_coordinates_spherical; optotagging:optotagging/optogenetic_stimulation; raw_gaze_mapping:eye_area/pupil_area/screen_coordinates/screen_coordinates_spherical; running:running_speed/running_speed_end_times/running_wheel_rotation; stimulus:timestamps
sub-775876808_ses-798911424	raw_running_wheel_rotation, running_wheel_signal_voltage, running_wheel_supply_voltage	eye_tracking:cr_ellipse_fits/eye_ellipse_fits/pupil_ellipse_fits; eye_tracking_rig_metadata:eye_tracking_rig_metadata; filtered_gaze_mapping:eye_area/pupil_area/screen_coordinates/screen_coordinates_spherical; optotagging:optotagging/optogenetic_stimulation; raw_gaze_mapping:eye_area/pupil_area/screen_coordinates/screen_coordinates_spherical; running:running_speed/running_speed_end_times/running_wheel_rotation; stimulus:timestamps
sub-776061236_ses-797828357	raw_running_wheel_rotation, running_wheel_signal_voltage, running_wheel_supply_voltage	eye_tracking:cr_ellipse_fits/eye_ellipse_fits/pupil_ellipse_fits; eye_tracking_rig_metadata:eye_tracking_rig_metadata; filtered_gaze_mapping:eye_area/pupil_area/screen_coordinates/screen_coordinates_spherical; optotagging:optotagging/optogenetic_stimulation; raw_gaze_mapping:eye_area/pupil_area/screen_coordinates/screen_coordinates_spherical; running:running_speed/running_speed_end_times/running_wheel_rotation; stimulus:timestamps

4. Units per region (Computed by this export)



5. All region labels (Computed by this export)

One row per region label exactly as stored in the NWB files, all units (no quality filter), sorted by unit count. Firing-rate columns summarise the NWB units.firing_rate values where the dataset provides them.

Region label	Units	Files	Subjects	firing_rate mean (Hz)	firing_rate median (Hz)
grey	13370	15	15	5.39	3.15
CA1	11644	26	26	4.01	2.47
VISp	5949	32	32	4.04	2.52
VISrl	4502	28	28	4.53	2.58
VISal	4254	24	24	3.92	2.50
DG	4086	25	25	3.93	2.14
VISam	3745	26	26	4.43	2.68
VISI	3517	24	24	3.97	2.71
LP	3497	22	22	6.29	5.02
VISpm	2527	20	20	4.05	2.71
APN	2392	22	22	8.40	3.86
CA3	2256	25	25	4.51	2.20
LGd	2179	21	21	7.81	6.67
VIS	1715	8	8	3.66	2.17
SUB	1606	15	15	6.56	4.36
(none)	759	27	27	3.76	1.82
ProS	726	12	12	4.73	2.94
PO	622	8	8	6.60	5.68
VPM	571	6	6	7.80	6.60
Eth	463	10	10	6.51	5.51
LGv	446	9	9	8.27	4.60
MGv	414	5	5	6.30	5.19
SGN	396	7	7	6.69	5.38
TH	384	19	19	6.31	3.79
VISmma	322	3	3	3.93	2.34
MB	305	15	15	5.90	2.24
POL	303	12	12	6.93	3.62
NOT	215	8	8	7.83	4.81
VISmmp	211	2	2	3.05	1.70
MGd	188	4	4	8.39	8.04
IGL	153	5	5	8.31	4.70
VISli	131	1	1	3.45	2.31
CA2	116	13	13	4.48	2.80
MGm	96	3	3	4.62	2.61
ZI	85	4	4	3.62	2.46
IntG	84	2	2	4.91	2.90
PPT	77	7	7	10.75	5.15
VL	63	1	1	3.94	1.94
SCig	60	4	4	5.78	3.57
PoT	49	3	3	8.85	8.66

Region label	Units	Files	Subjects	firing_rate mean (Hz)	firing_rate median (Hz)
PP	46	2	2	10.30	6.94
OP	40	2	2	9.92	4.85
PIL	38	2	2	5.33	4.18
MRN	18	2	2	13.30	8.84
COAa	16	1	1	12.07	4.70
VPL	11	1	1	5.11	3.91
HPF	7	2	2	1.30	0.95
LD	3	1	1	3.09	2.84
POST	1	1	1	3.96	3.96

6. Intervals tables (stimuli, trials, epochs)

Each NWB intervals table, with the number of files containing it, total rows, and its columns as stored.

Table	Files	Rows	Columns
drifting_gratings_contrast_presentations	8	4,320	start_time, stop_time, stimulus_name, stimulus_block, temporal_frequency, color, mask, opacity, phase, size, units, stimulus_index, orientation, spatial_frequency, contrast
drifting_gratings_presentations	32	20,152	start_time, stop_time, stimulus_name, stimulus_block, temporal_frequency, color, mask, opacity, phase, size, units, stimulus_index, orientation, spatial_frequency, contrast
flashes_presentations	32	4,800	start_time, stop_time, stimulus_name, stimulus_block, color, mask, opacity, phase, size, units, stimulus_index, orientation, spatial_frequency, contrast
gabor_presentations	32	116,640	start_time, stop_time, stimulus_name, stimulus_block, temporal_frequency, x_position, y_position, color, mask, opacity, phase, size, units, stimulus_index, orientation, spatial_frequency, contrast
invalid_times	13	54	start_time, stop_time
natural_movie_one_presentations	32	576,000	start_time, stop_time, stimulus_name, stimulus_block, color, opacity, size, units, stimulus_index, orientation, frame, contrast
natural_movie_three_presentations	32	1,152,000	start_time, stop_time, stimulus_name, stimulus_block, color, opacity, size, units, stimulus_index, orientation, frame, contrast
natural_scenes_presentations	32	190,400	start_time, stop_time, stimulus_name, stimulus_block, stimulus_index, frame
spontaneous_presentations	32	488	start_time, stop_time, stimulus_name
static_gratings_presentations	32	192,000	start_time, stop_time, stimulus_name, stimulus_block, color, mask, opacity, phase, size, units, stimulus_index, orientation, spatial_frequency, contrast

7. Column and signal descriptions from the NWB files

Quoted verbatim from sub-699733573_ses-715093703.nwb; all files in the Dandiset share this structure.

Location in NWB file	Name	Description (as written in the NWB file)
electrodes	(table)	metadata about extracellular electrodes
electrodes	x	the x coordinate of the channel location
electrodes	y	the y coordinate of the channel location
electrodes	z	the z coordinate of the channel location
electrodes	imp	the impedance of the channel
electrodes	location	the location of channel within the subject e.g. brain region
electrodes	filtering	description of hardware filtering
electrodes	group	a reference to the ElectrodeGroup this electrode is a part of
electrodes	group_name	the name of the ElectrodeGroup this electrode is a part of
electrodes	probe_vertical_position	Length-wise position of electrode/channel on device (microns)
electrodes	probe_horizontal_position	Width-wise position of electrode/channel on device (microns)
electrodes	probe_id	The unique id of this electrode's/channel's device
electrodes	valid_data	Whether data from this electrode/channel is usable
units	(table)	
units	snr	no description
units	quality	no description
units	presence_ratio	no description
units	spread	no description
units	cumulative_drift	no description
units	isolation_distance	no description
units	amplitude	no description
units	firing_rate	no description
units	nn_hit_rate	no description
units	isi_violations	no description
units	velocity_below	no description
units	l_ratio	no description
units	nn_miss_rate	no description
units	waveform_halfwidth	no description
units	repolarization_slope	no description
units	amplitude_cutoff	no description
units	PT_ratio	no description
units	silhouette_score	no description
units	recovery_slope	no description
units	waveform_duration	no description
units	max_drift	no description
units	velocity_above	no description
units	cluster_id	no description
units	peak_channel_id	no description
units	d_prime	no description
units	spike_times	times (s) of detected spiking events

Location in NWB file	Name	Description (as written in the NWB file)
units	spike_amplitudes	amplitude (s) of detected spiking events
units	waveform_mean	mean waveforms on peak channels (and over samples)
intervals/drifted_gratings_presentations	(table)	Presentation times and stimuli details for 'drifted_gratings' stimuli
intervals/drifted_gratings_presentations	start_time	Start time of epoch, in seconds
intervals/drifted_gratings_presentations	stop_time	Stop time of epoch, in seconds
intervals/drifted_gratings_presentations	stimulus_name	Name of stimulus
intervals/drifted_gratings_presentations	stimulus_block	Index of contiguous presentations of one stimulus type
intervals/drifted_gratings_presentations	temporal_frequency	Temporal frequency of stimulus
intervals/drifted_gratings_presentations	color	No description
intervals/drifted_gratings_presentations	mask	Shape of mask applied to stimulus
intervals/drifted_gratings_presentations	opacity	Opacity of stimulus
intervals/drifted_gratings_presentations	phase	Phase of grating stimulus
intervals/drifted_gratings_presentations	size	Size of stimulus (see 'units' field for units)
intervals/drifted_gratings_presentations	units	Units of stimulus size
intervals/drifted_gratings_presentations	orientation	Orientation of stimulus
intervals/drifted_gratings_presentations	spatial_frequency	Spatial frequency of stimulus
intervals/drifted_gratings_presentations	contrast	Contrast of stimulus
intervals/drifted_gratings_presentations	tags	user-defined tags
intervals/drifted_gratings_presentations	timeseries	index into a TimeSeries object
intervals/flushes_presentations	(table)	Presentation times and stimuli details for 'flushes' stimuli
intervals/flushes_presentations	start_time	Start time of epoch, in seconds
intervals/flushes_presentations	stop_time	Stop time of epoch, in seconds
intervals/flushes_presentations	stimulus_name	Name of stimulus
intervals/flushes_presentations	stimulus_block	Index of contiguous presentations of one stimulus type
intervals/flushes_presentations	color	No description
intervals/flushes_presentations	mask	Shape of mask applied to stimulus
intervals/flushes_presentations	opacity	Opacity of stimulus
intervals/flushes_presentations	phase	Phase of grating stimulus
intervals/flushes_presentations	size	Size of stimulus (see 'units' field for units)
intervals/flushes_presentations	units	Units of stimulus size
intervals/flushes_presentations	orientation	Orientation of stimulus
intervals/flushes_presentations	spatial_frequency	Spatial frequency of stimulus
intervals/flushes_presentations	contrast	Contrast of stimulus
intervals/flushes_presentations	tags	user-defined tags
intervals/flushes_presentations	timeseries	index into a TimeSeries object
intervals/gabors_presentations	(table)	Presentation times and stimuli details for 'gabors' stimuli
intervals/gabors_presentations	start_time	Start time of epoch, in seconds
intervals/gabors_presentations	stop_time	Stop time of epoch, in seconds
intervals/gabors_presentations	stimulus_name	Name of stimulus
intervals/gabors_presentations	stimulus_block	Index of contiguous presentations of one stimulus type
intervals/gabors_presentations	temporal_frequency	Temporal frequency of stimulus
intervals/gabors_presentations	x_position	Horizontal position of stimulus on screen
intervals/gabors_presentations	y_position	Vertical position of stimulus on screen

Location in NWB file	Name	Description (as written in the NWB file)
intervals/gabors_presentations	color	No description
intervals/gabors_presentations	mask	Shape of mask applied to stimulus
intervals/gabors_presentations	opacity	Opacity of stimulus
intervals/gabors_presentations	phase	Phase of grating stimulus
intervals/gabors_presentations	size	Size of stimulus (see 'units' field for units)
intervals/gabors_presentations	units	Units of stimulus size
intervals/gabors_presentations	orientation	Orientation of stimulus
intervals/gabors_presentations	spatial_frequency	Spatial frequency of stimulus
intervals/gabors_presentations	contrast	Contrast of stimulus
intervals/gabors_presentations	tags	user-defined tags
intervals/gabors_presentations	timeseries	index into a TimeSeries object
intervals/invalid_times	(table)	experimental intervals
intervals/invalid_times	start_time	Start time of epoch, in seconds
intervals/invalid_times	stop_time	Stop time of epoch, in seconds
intervals/invalid_times	tags	user-defined tags
intervals/natural_movie_one_presentations	(table)	Presentation times and stimuli details for 'natural_movie_one' stimuli
intervals/natural_movie_one_presentations	start_time	Start time of epoch, in seconds
intervals/natural_movie_one_presentations	stop_time	Stop time of epoch, in seconds
intervals/natural_movie_one_presentations	stimulus_name	Name of stimulus
intervals/natural_movie_one_presentations	stimulus_block	Index of contiguous presentations of one stimulus type
intervals/natural_movie_one_presentations	color	No description
intervals/natural_movie_one_presentations	opacity	Opacity of stimulus
intervals/natural_movie_one_presentations	size	Size of stimulus (see 'units' field for units)
intervals/natural_movie_one_presentations	units	Units of stimulus size
intervals/natural_movie_one_presentations	orientation	Orientation of stimulus
intervals/natural_movie_one_presentations	frame	Frame of movie stimulus
intervals/natural_movie_one_presentations	contrast	Contrast of stimulus
intervals/natural_movie_one_presentations	tags	user-defined tags
intervals/natural_movie_one_presentations	timeseries	index into a TimeSeries object
intervals/natural_movie_three_presentations	(table)	Presentation times and stimuli details for 'natural_movie_three' stimuli
intervals/natural_movie_three_presentations	start_time	Start time of epoch, in seconds

Location in NWB file	Name	Description (as written in the NWB file)
intervals/natural_movie_three_presentations	stop_time	Stop time of epoch, in seconds
intervals/natural_movie_three_presentations	stimulus_name	Name of stimulus
intervals/natural_movie_three_presentations	stimulus_block	Index of contiguous presentations of one stimulus type
intervals/natural_movie_three_presentations	color	No description
intervals/natural_movie_three_presentations	opacity	Opacity of stimulus
intervals/natural_movie_three_presentations	size	Size of stimulus (see 'units' field for units)
intervals/natural_movie_three_presentations	units	Units of stimulus size
intervals/natural_movie_three_presentations	orientation	Orientation of stimulus
intervals/natural_movie_three_presentations	frame	Frame of movie stimulus
intervals/natural_movie_three_presentations	contrast	Contrast of stimulus
intervals/natural_movie_three_presentations	tags	user-defined tags
intervals/natural_movie_three_presentations	timeseries	index into a TimeSeries object
intervals/natural_scenes_presentations	(table)	Presentation times and stimuli details for 'natural_scenes' stimuli
intervals/natural_scenes_presentations	start_time	Start time of epoch, in seconds
intervals/natural_scenes_presentations	stop_time	Stop time of epoch, in seconds
intervals/natural_scenes_presentations	stimulus_name	Name of stimulus
intervals/natural_scenes_presentations	stimulus_block	Index of contiguous presentations of one stimulus type
intervals/natural_scenes_presentations	frame	Frame of movie stimulus
intervals/natural_scenes_presentations	tags	user-defined tags
intervals/natural_scenes_presentations	timeseries	index into a TimeSeries object
intervals/spontaneous_presentations	(table)	Presentation times and stimuli details for 'spontaneous' stimuli
intervals/spontaneous_presentations	start_time	Start time of epoch, in seconds
intervals/spontaneous_presentations	stop_time	Stop time of epoch, in seconds
intervals/spontaneous_presentations	stimulus_name	Name of stimulus
intervals/spontaneous_presentations	tags	user-defined tags
intervals/spontaneous_presentations	timeseries	index into a TimeSeries object
intervals/static_gratings_presentations	(table)	Presentation times and stimuli details for 'static_gratings' stimuli
intervals/static_gratings_presentations	start_time	Start time of epoch, in seconds
intervals/static_gratings_presentations	stop_time	Stop time of epoch, in seconds
intervals/static_gratings_presentations	stimulus_name	Name of stimulus
intervals/static_gratings_presentations	stimulus_block	Index of contiguous presentations of one stimulus type
intervals/static_gratings_presentations	color	No description
intervals/static_gratings_presentations	mask	Shape of mask applied to stimulus
intervals/static_gratings_presentations	opacity	Opacity of stimulus

Location in NWB file	Name	Description (as written in the NWB file)
intervals/static_gratings_presentations	phase	Phase of grating stimulus
intervals/static_gratings_presentations	size	Size of stimulus (see 'units' field for units)
intervals/static_gratings_presentations	units	Units of stimulus size
intervals/static_gratings_presentations	orientation	Orientation of stimulus
intervals/static_gratings_presentations	spatial_frequency	Spatial frequency of stimulus
intervals/static_gratings_presentations	contrast	Contrast of stimulus
intervals/static_gratings_presentations	tags	user-defined tags
intervals/static_gratings_presentations	timeseries	index into a TimeSeries object
acquisition	raw_running_wheel_rotation	no description
acquisition	running_wheel_signal_voltage	no description
acquisition	running_wheel_supply_voltage	no description
processing/eye_tracking_rig_metadata	(module)	Eye tracking rig metadata module
processing/eye_tracking_rig_metadata	eye_tracking_rig_metadata	
processing/optotagging	(module)	optogenetic stimulation data
processing/optotagging	optotagging	no description
processing/optotagging/optogenetic_stimulation	(table)	
processing/optotagging/optogenetic_stimulation	start_time	Start time of epoch, in seconds
processing/optotagging/optogenetic_stimulation	condition	no description
processing/optotagging/optogenetic_stimulation	level	no description
processing/optotagging/optogenetic_stimulation	stop_time	Stop time of epoch, in seconds
processing/optotagging/optogenetic_stimulation	stimulus_name	no description
processing/optotagging/optogenetic_stimulation	duration	no description
processing/optotagging/optogenetic_stimulation	tags	user-defined tags
processing/optotagging/optogenetic_stimulation	timeseries	index into a TimeSeries object
processing/running	(module)	running speed data
processing/running	running_speed	no description
processing/running	running_speed_end_times	no description
processing/running	running_wheel_rotation	no description
processing/stimulus	(module)	Stimulus Times processing
processing/stimulus	timestamps	no description

8. Exported files

File	Rows	Columns
tables/sessions.csv	32	session_id, asset_path, size_gb, subject_id, species, sex, age, genotype, strain, session_start_time, session_description, institution, lab, n_units, n_electrodes, electrode_groups, region_source, regions, intervals, acquisition, processing
tables/units.parquet	74,658	session_id, unit_id, unit_row, region, snr, quality, presence_ratio, spread, cumulative_drift, isolation_distance, amplitude, firing_rate, nn_hit_rate, isi_violations, velocity_below, l_ratio, nn_miss_rate, waveform_halfwidth, repolarization_slope, amplitude_cutoff, PT_ratio, silhouette_score, recovery_slope, waveform_duration, max_drift, velocity_above, cluster_id, peak_channel_id, d_prime, subject_id
tables/region_summary.csv	49	region, n_units, n_sessions, n_subjects, firing_rate_mean_hz, firing_rate_median_hz
tables/intervals_summary.csv	10	table, n_sessions, n_rows, columns
intervals/<table>.parquet	one per intervals table	session_id + the table's columns
tables/dataset_metadata.json, nwb_file_metadata.json, nwb_descriptions.csv		published metadata quoted in sections 1 and 7